

Contents

1	Mathematical Foundations of GAMLSS	1
1.1	1. Distribution Derivatives and Distributional Functions	1
1.2	2. Special Functions	11
1.3	3. Numerical Stability and Implementation	12
1.4	4. GAMLSS Iteration (P-IRLS)	13
1.5	5. Smoothing and Penalty Matrices	16
1.6	6. Effective Degrees of Freedom	20
1.7	7. Link Functions	20
1.8	8. GCV Gradient for Smoothing Parameter Optimization	21
1.9	9. Mean-Variance Relationships	26
1.10	10. Convergence and Initialization	27
1.11	11. Model Selection and Diagnostics	28
1.12	12. Code Map	30
1.13	13. Notation Glossary	31
1.14	References	32

1 Mathematical Foundations of GAMLSS

This document provides detailed mathematical derivations for the GAMLSS implementation in **glissando**. It covers distribution-specific derivatives, special functions, numerical algorithms, and convergence theory.

Generated PDF: run `bash docs/math/build-math-pdf.sh` to produce `docs/math/mathematics.pdf` (a numbered, hyperlinked, table-of-contents'd PDF). The PDF build uses pandoc + xelatex; see the script header for details.

1.1 1. Distribution Derivatives and Distributional Functions

This section provides detailed derivations of score functions (first derivatives of log-likelihood) and Fisher information (expected second derivatives) for all implemented distributions. These quantities drive the penalized iteratively reweighted least squares (P-IRLS) algorithm. §1.10 then derives the cumulative distribution, density, and quantile (the CDF/PDF/quantile trio) that these same families expose for residuals, centiles, and predictive intervals.

Key quantities: - **Score** $u = \frac{\partial \ell}{\partial \eta}$: Direction of steepest ascent for the log-likelihood - **Fisher information** $w = I_{\eta}$: Expected curvature (used as weight in IRLS) - **Working response** $z = \eta + u/w$: Adjusted target for the weighted least squares problem

For each distribution, we derive these quantities accounting for the link function via the chain rule.

1.1.1 1.1 Gaussian Distribution

Parameterization: $Y \sim N(\mu, \sigma^2)$

Parameters: - μ (mean): identity link - σ (standard deviation): log link

Variance: $\text{Var}(Y) = \sigma^2$

1.1.1.1 Log-Likelihood

$$\ell(\mu, \sigma | y) = -\frac{1}{2} \log(2\pi) - \log(\sigma) - \frac{(y - \mu)^2}{2\sigma^2}$$

1.1.1.2 Derivatives for μ (identity link) Score:

$$\frac{\partial \ell}{\partial \mu} = \frac{y - \mu}{\sigma^2}$$

Fisher information:

$$I_\mu = \mathbb{E} \left[-\frac{\partial^2 \ell}{\partial \mu^2} \right] = \frac{1}{\sigma^2}$$

Since we use identity link ($\eta = \mu$), no chain rule needed:

$$u_\mu = \frac{y - \mu}{\sigma^2}, \quad w_\mu = \frac{1}{\sigma^2}$$

1.1.1.3 Derivatives for σ (log link) Let $\eta_\sigma = \log(\sigma)$, so $\sigma = e^{\eta_\sigma}$.

Score with respect to σ :

$$\frac{\partial \ell}{\partial \sigma} = -\frac{1}{\sigma} + \frac{(y - \mu)^2}{\sigma^3}$$

Chain rule for log link:

$$\frac{\partial \ell}{\partial \eta_\sigma} = \sigma \cdot \frac{\partial \ell}{\partial \sigma} = -1 + \frac{(y - \mu)^2}{\sigma^2} = \frac{(y - \mu)^2 - \sigma^2}{\sigma^2}$$

Fisher information on η_σ scale:

$$I_{\eta_\sigma} = \mathbb{E} \left[-\frac{\partial^2 \ell}{\partial \eta_\sigma^2} \right] = 2$$

This comes from $\mathbb{E}[(Y - \mu)^2/\sigma^2] = 1$ and $\text{Var}[(Y - \mu)^2/\sigma^2] = 2$ for Gaussian.

Implementation:

$$u_\sigma = \frac{(y - \mu)^2 - \sigma^2}{\sigma^2}, \quad w_\sigma = 2$$

1.1.2 1.2 Student-t Distribution

1.1.3 Log-Likelihood

For observation y with parameters μ (location), σ (scale), and ν (degrees of freedom):

$$\ell(\mu, \sigma, \nu | y) = \log \Gamma \left(\frac{\nu + 1}{2} \right) - \log \Gamma \left(\frac{\nu}{2} \right) - \frac{1}{2} \log(\nu \pi \sigma^2) - \frac{\nu + 1}{2} \log \left(1 + \frac{z^2}{\nu} \right)$$

where $z = \frac{y - \mu}{\sigma}$.

1.1.4 First Derivatives (Score Functions)

Define the robustified weight:

$$w = \frac{\nu + 1}{\nu + z^2}$$

1.1.4.1 Location parameter μ :

$$\frac{\partial \ell}{\partial \mu} = \frac{w \cdot z}{\sigma}$$

1.1.4.2 Scale parameter σ (log link):

$$\frac{\partial \ell}{\partial \log(\sigma)} = w \cdot z^2 - 1$$

1.1.4.3 Degrees of freedom ν (log link):

$$\frac{\partial \ell}{\partial \log(\nu)} = \nu \left[\frac{1}{2} \left(\psi \left(\frac{\nu+1}{2} \right) - \psi \left(\frac{\nu}{2} \right) - \log \left(1 + \frac{z^2}{\nu} \right) + \frac{w \cdot z^2 - 1}{\nu} \right) \right]$$

where $\psi(x) = \frac{d}{dx} \log \Gamma(x)$ is the digamma function.

1.1.5 Expected Fisher Information

1.1.5.1 For μ :

$$I_\mu = \mathbb{E} \left[\frac{w}{\sigma^2} \right] = \frac{\nu + 1}{\sigma^2(\nu + 3)}$$

The working weight uses this expected information (the same convention as `gamlss TF()`), not the observed per-row weight w/σ^2 : the two agree only at $z^2 = 3$, and a data-dependent weight changes the PWLS subproblem — hence λ selection, EDF, and standard errors — relative to the reference implementation.

1.1.5.2 For σ (log link):

$$I_{\log(\sigma)} = \frac{2\nu}{\nu + 3}$$

1.1.5.3 For ν (log link) Lange, Little & Taylor (1989), identical to `gamlss TF()`'s `d21dv2`:

$$I_{\log(\nu)} = \frac{\nu^2}{4} \left[\psi' \left(\frac{\nu}{2} \right) - \psi' \left(\frac{\nu+1}{2} \right) - \frac{2(\nu+5)}{\nu(\nu+1)(\nu+3)} \right]$$

where $\psi'(x)$ is the trigamma function. The rational term is a small correction: $I_{\log(\nu)}$ decays like $O(1/\nu)$, so a sign or degree error here inflates the weight by orders of magnitude and effectively freezes ν at its starting value (an earlier implementation used $+2(\nu+3)/(\nu(\nu+1))$, $\sim 50\times$ too large at $\nu = 5$). The formula is validated against a Monte-Carlo estimate of $\mathbb{E}[(\nu \partial \ell / \partial \nu)^2]$.

1.1.6 Numerical Considerations

1. **Minimum ν :** $\nu > 2$ is enforced to ensure finite variance, via a *floored log link* on ν ($\eta = \log \nu$, $\nu = \max(e^\eta, 2)$). The optimizer can explore the heavy-tail region without the variance $\sigma^2 \nu / (\nu - 2)$ becoming undefined; the floor never binds when the true ν is well above 2. See `FlooredLogLink` in `src/distributions/links.rs`.

Where the floor binds, $d\nu/d\eta = 0$ and the score must not be forwarded blindly. The implementation applies a KKT-style aggregate projection: if the score summed over the pinned rows is ≤ 0 , the constrained optimum is at the boundary and those rows are frozen ($u = 0$, so the block reports a zero step and the outer loop converges); if it is > 0 the full chain rule is forwarded so ν can re-enter the interior. Forwarding a negative aggregate walks η_ν downward indefinitely; a per-row one-sided projection instead biases the aggregate upward and produces a lift-off / fall-back limit cycle at the boundary. The single summed score is exact for an intercept-only η_ν (the standard usage).

2. **Minimum σ :** Enforce $\sigma \geq 10^{-6}$ to prevent division by zero
3. **Weight clamping:** The denominator $\nu + z^2$ should be $\geq 10^{-10}$
4. **Information positivity:** Ensure $I_{\log(\nu)} \geq 10^{-6}$

1.1.7 1.3 Poisson Distribution

Parameterization: $Y \sim \text{Poisson}(\mu)$

Parameters: - μ (rate/mean): log link

Variance: $\text{Var}(Y) = \mu$ (equidispersion)

1.1.7.1 Log-Likelihood

$$\ell(\mu|y) = y \log(\mu) - \mu - \log(y!)$$

1.1.7.2 Derivatives for μ (log link) Let $\eta = \log(\mu)$, so $\mu = e^\eta$.

Score with respect to μ :

$$\frac{\partial \ell}{\partial \mu} = \frac{y}{\mu} - 1 = \frac{y - \mu}{\mu}$$

Chain rule for log link:

$$\frac{\partial \ell}{\partial \eta} = \mu \cdot \frac{\partial \ell}{\partial \mu} = y - \mu$$

Fisher information on η scale:

$$I_\eta = \mathbb{E} \left[-\frac{\partial^2 \ell}{\partial \eta^2} \right] = \mu$$

Implementation:

$$u_\mu = y - \mu, \quad w_\mu = \mu$$

1.1.8 1.4 Gamma Distribution

Parameterization: $Y \sim \text{Gamma}(\alpha, \theta)$ where $\alpha = 1/\sigma^2$ (shape), $\theta = \mu\sigma^2$ (scale)

Parameters: - μ (mean): log link - σ (coefficient of variation): log link

Variance: $\text{Var}(Y) = \mu^2 \sigma^2$

Mean-CV relationship: $\text{CV}(Y) = \sigma$

1.1.8.1 Log-Likelihood

$$\ell(\mu, \sigma|y) = -\alpha \log(\theta) - \log \Gamma(\alpha) + (\alpha - 1) \log(y) - \frac{y}{\theta}$$

where $\alpha = 1/\sigma^2$ and $\theta = \mu\sigma^2$.

Simplifying:

$$\ell = -\frac{1}{\sigma^2} \log(\mu\sigma^2) - \log \Gamma\left(\frac{1}{\sigma^2}\right) + \left(\frac{1}{\sigma^2} - 1\right) \log(y) - \frac{y}{\mu\sigma^2}$$

1.1.8.2 Derivatives for μ (log link) Let $\eta_\mu = \log(\mu)$.

Score with respect to μ :

$$\frac{\partial \ell}{\partial \mu} = -\frac{\alpha}{\mu} + \frac{y}{\mu^2 \theta} = -\frac{1}{\mu\sigma^2} + \frac{y}{\mu^2 \sigma^2} = \frac{y - \mu}{\mu^2 \sigma^2}$$

Chain rule for log link:

$$\frac{\partial \ell}{\partial \eta_\mu} = \mu \cdot \frac{\partial \ell}{\partial \mu} = \frac{y - \mu}{\mu \sigma^2}$$

Fisher information on η_μ scale:

$$I_{\eta_\mu} = \frac{1}{\sigma^2}$$

Implementation:

$$u_\mu = \frac{y - \mu}{\mu\sigma^2}, \quad w_\mu = \frac{1}{\sigma^2}$$

1.1.8.3 Derivatives for σ (log link) Let $\eta_\sigma = \log(\sigma)$.

Score with respect to σ :

$$\frac{\partial \ell}{\partial \sigma} = -\frac{2\alpha}{\sigma} + \frac{2\alpha}{\sigma} \psi(\alpha) + \frac{2\alpha}{\sigma} \log(y) - \frac{2\alpha}{\sigma} \log(\theta) + \frac{2y}{\mu\sigma^3}$$

where $\psi(x) = \frac{d}{dx} \log \Gamma(x)$ is the digamma function.

Chain rule for log link ($\frac{d\sigma}{d\eta_\sigma} = \sigma$):

$$\frac{\partial \ell}{\partial \eta_\sigma} = \frac{2}{\sigma^2} \left[\psi\left(\frac{1}{\sigma^2}\right) + 2\log(\sigma) - \log\left(\frac{y}{\mu}\right) + \frac{y}{\mu} - 1 \right]$$

Fisher information involves trigamma $\psi'(x) = \frac{d^2}{dx^2} \log \Gamma(x)$. On the natural scale $I_\sigma = \frac{4}{\sigma^6} \psi'(\alpha) - \frac{4}{\sigma^4}$, so on the η_σ scale ($\times \sigma^2$):

$$I_{\eta_\sigma} = \frac{4}{\sigma^4} \psi'\left(\frac{1}{\sigma^2}\right) - \frac{4}{\sigma^2}$$

This matches gamlss **GA**'s **d2l1dd2** and the Monte-Carlo check $\mathbb{E}[u_{\eta_\sigma}^2]$. Since $\psi'(1/\sigma^2) > \sigma^2$ for all $\sigma > 0$, the expression is strictly positive, with limit $I_{\eta_\sigma} \rightarrow 2$ as $\sigma \rightarrow 0$ (the Gaussian-like regime). An earlier implementation used $-2/\sigma^2$ for the second term, inflating the weight roughly five-fold at $\sigma = 0.5$ and over-damping every σ update.

Implementation:

$$u_\sigma = \frac{2}{\sigma^2} [\psi(\alpha) + 2\log(\sigma) - \log(y/\mu) + y/\mu - 1], \quad w_\sigma = \frac{4}{\sigma^4} \psi'(\alpha) - \frac{4}{\sigma^2}$$

1.1.9 1.5 Negative Binomial Distribution

Parameterization: $Y \sim \text{NB}(r, p)$ with mean μ and dispersion σ (NB2 parameterization)

Parameters: - μ (mean): log link - σ (overdispersion): log link

Variance: $\text{Var}(Y) = \mu + \sigma\mu^2$ (overdispersion relative to Poisson)

Relationship to standard NB: $r = 1/\sigma$, $p = 1/(1 + \sigma\mu)$

1.1.9.1 Log-Likelihood

$$\ell(\mu, \sigma | y) = \log \Gamma(y + r) - \log \Gamma(r) - \log(y!) + r \log(p) + y \log(1 - p)$$

where $r = 1/\sigma$ and $p = 1/(1 + \sigma\mu)$.

Simplifying in terms of (μ, σ) :

$$\ell = \log \Gamma(y + 1/\sigma) - \log \Gamma(1/\sigma) - \log(y!) + \frac{1}{\sigma} \log\left(\frac{1}{1 + \sigma\mu}\right) + y \log\left(\frac{\sigma\mu}{1 + \sigma\mu}\right)$$

1.1.9.2 Derivatives for μ (log link)

Let $\eta_\mu = \log(\mu)$.

Score with respect to μ :

$$\frac{\partial \ell}{\partial \mu} = \frac{y - \mu}{1 + \sigma \mu}$$

Chain rule for log link ($\frac{d\mu}{d\eta_\mu} = \mu$):

$$\frac{\partial \ell}{\partial \eta_\mu} = \mu \cdot \frac{\partial \ell}{\partial \mu} = \frac{\mu(y - \mu)}{1 + \sigma \mu}$$

However, in IRLS we use the observed Fisher information $w_\mu = \frac{\mu}{1 + \sigma \mu}$, giving working response:

$$u_\mu = \frac{y - \mu}{1 + \sigma \mu}, \quad w_\mu = \frac{\mu}{1 + \sigma \mu}$$

This follows from the score being proportional to observed weight, as noted in the full derivation in Rigby & Stasinopoulos (2005).

Why observed information here: for μ in NB the expected information $I_{\eta_\mu} = \mathbb{E}[\mu(Y - \mu)/(1 + \sigma \mu)^2]$ collapses to $\mu/(1 + \sigma \mu)$ after expectation, which coincides numerically with the observed-information form because of the variance identity $\text{Var}(Y) = \mu(1 + \sigma \mu)$. Rigby & Stasinopoulos (2005, eq. (15)) use this simplification; `src/distributions/negative_binomial.rs` follows it. As $\sigma \rightarrow 0$ the weight reduces to μ — exactly the Poisson Fisher weight — confirming the limiting Poisson behavior.

Implementation (for IRLS):

$$u_\mu = \frac{y - \mu}{1 + \sigma \mu}, \quad w_\mu = \frac{\mu}{1 + \sigma \mu}$$

1.1.9.3 Derivatives for σ (log link)

Let $\eta_\sigma = \log(\sigma)$ and $r = 1/\sigma$.

Differentiating with respect to r first,

$$\frac{\partial \ell}{\partial r} = \psi(y + r) - \psi(r) - \log(1 + \sigma \mu) + \frac{\mu - y}{r + \mu},$$

then $\frac{\partial \ell}{\partial \eta_\sigma} = \sigma \frac{\partial \ell}{\partial \sigma} = -\frac{1}{\sigma} \frac{\partial \ell}{\partial r}$ (since $r = 1/\sigma$):

$$\frac{\partial \ell}{\partial \eta_\sigma} = -\frac{1}{\sigma} \left[\psi(y + r) - \psi(r) - \log(1 + \sigma \mu) + \frac{\mu - y}{r + \mu} \right]$$

Working weight — the exact expected information has no closed form (it involves $\mathbb{E}[\psi'(Y + r)]$), so the squared-score convention of gamlss NBI (`d2l1dd2 = -d1dd^2`) is used:

$$w_{\eta_\sigma} = u_{\eta_\sigma}^2 \quad (\text{floored at } 10^{-6})$$

An earlier implementation used the partial expected-information approximation $\psi'(r)/\sigma^2$, which drops same-order terms, behaves like $1/\sigma$ near the Poisson boundary (over-damping σ updates), and does not track the gamlss oracle's λ/EDF selection for σ smooths.

Implementation:

$$u_\sigma = -\frac{1}{\sigma} \left[\psi(y + 1/\sigma) - \psi(1/\sigma) - \log(1 + \sigma \mu) + \frac{\mu - y}{1/\sigma + \mu} \right], \quad w_\sigma = \max(u_\sigma^2, 10^{-6})$$

1.1.10 1.6 Beta Distribution

Parameterization: $Y \sim \text{Beta}(\alpha, \beta)$ where $\alpha = \mu\phi$ and $\beta = (1 - \mu)\phi$

Parameters: - μ (mean): logit link - ϕ (precision): log link

Variance: $\text{Var}(Y) = \frac{\mu(1-\mu)}{1+\phi}$

1.1.10.1 Log-Likelihood

$$\ell(\mu, \phi|y) = \log \Gamma(\phi) - \log \Gamma(\alpha) - \log \Gamma(\beta) + (\alpha - 1) \log(y) + (\beta - 1) \log(1 - y)$$

where $\alpha = \mu\phi$ and $\beta = (1 - \mu)\phi$.

1.1.10.2 Derivatives for μ (logit link) Let $\eta_\mu = \text{logit}(\mu) = \log(\mu/(1 - \mu))$.

Score with respect to μ :

$$\frac{\partial \ell}{\partial \mu} = \phi [\log(y) - \log(1 - y) - \psi(\alpha) + \psi(\beta)]$$

Chain rule for logit link ($\frac{d\mu}{d\eta_\mu} = \mu(1 - \mu)$):

$$\frac{\partial \ell}{\partial \eta_\mu} = \mu(1 - \mu) \cdot \frac{\partial \ell}{\partial \mu} = \mu(1 - \mu)\phi [\log(y) - \log(1 - y) - \psi(\alpha) + \psi(\beta)]$$

Fisher information:

$$I_{\eta_\mu} = [\mu(1 - \mu)]^2 \phi^2 [\psi'(\alpha) + \psi'(\beta)]$$

Implementation:

$$\begin{aligned} u_\mu &= \mu(1 - \mu)\phi [\log(y) - \log(1 - y) - \psi(\mu\phi) + \psi((1 - \mu)\phi)] \\ w_\mu &= [\mu(1 - \mu)]^2 \phi^2 [\psi'(\mu\phi) + \psi'((1 - \mu)\phi)] \end{aligned}$$

1.1.10.3 Derivatives for ϕ (log link) Let $\eta_\phi = \log(\phi)$.

Score with respect to ϕ :

$$\frac{\partial \ell}{\partial \phi} = \psi(\phi) - \mu\psi(\alpha) - (1 - \mu)\psi(\beta) + \mu \log(y) + (1 - \mu) \log(1 - y)$$

Chain rule for log link:

$$\frac{\partial \ell}{\partial \eta_\phi} = \phi \cdot \frac{\partial \ell}{\partial \phi}$$

Fisher information:

$$I_{\eta_\phi} = \phi^2 [\mu^2 \psi'(\alpha) + (1 - \mu)^2 \psi'(\beta) - \psi'(\phi)]$$

The bracket is $-\partial^2 \ell / \partial \phi^2$; the sign convention matters because ψ' is decreasing and convex, which makes $\mu^2 \psi'(\mu\phi) + (1 - \mu)^2 \psi'((1 - \mu)\phi) > \psi'(\phi)$ for all $\mu \in (0, 1)$, $\phi > 0$ — so I_{η_ϕ} is strictly positive as an information must be. (An earlier version wrote the bracket negated and relied on an absolute value.)

Implementation:

$$\begin{aligned} u_\phi &= \phi [\psi(\phi) - \mu\psi(\mu\phi) - (1 - \mu)\psi((1 - \mu)\phi) + \mu \log(y) + (1 - \mu) \log(1 - y)] \\ w_\phi &= \phi^2 [\mu^2 \psi'(\mu\phi) + (1 - \mu)^2 \psi'((1 - \mu)\phi) - \psi'(\phi)] \end{aligned}$$

1.1.11 1.7 Binomial Distribution

Parameterization: $Y \sim \text{Binomial}(n, \mu)$ where Y is the number of successes in n trials

Parameters: - μ (probability): logit link

Variance: $\text{Var}(Y) = n\mu(1 - \mu)$

1.1.11.1 Log-Likelihood

$$\ell(\mu|y) = y \log(\mu) + (n - y) \log(1 - \mu) + \log \binom{n}{y}$$

1.1.11.2 Derivatives for μ (logit link) Let $\eta = \text{logit}(\mu)$.

Score with respect to μ :

$$\frac{\partial \ell}{\partial \mu} = \frac{y}{\mu} - \frac{n - y}{1 - \mu} = \frac{y - n\mu}{\mu(1 - \mu)}$$

Chain rule for logit link ($\frac{d\mu}{d\eta} = \mu(1 - \mu)$):

$$\frac{\partial \ell}{\partial \eta} = \mu(1 - \mu) \cdot \frac{\partial \ell}{\partial \mu} = y - n\mu$$

Fisher information on η scale:

$$I_\eta = n\mu(1 - \mu)$$

Implementation:

$$u_\mu = y - n\mu, \quad w_\mu = n\mu(1 - \mu)$$

1.1.12 1.8 Ordered Categorical Distribution (0cat)

Parameterization: Proportional-odds / cumulative-logit for ordinal response $y \in \{1, \dots, R\}$, $R \in \{2, 3, 4, 5\}$.

Parameters: - μ (latent linear predictor): identity link - δ_1 (first threshold θ_1): identity link - δ_k (positive increment $\theta_k - \theta_{k-1}$), $k = 2, \dots, R - 1$: log link

Threshold reconstruction: $\theta_1 = \delta_1$ (unconstrained); $\theta_k = \theta_{k-1} + \exp(\delta_k)$ for $k \geq 2$, ensuring strict monotonicity $\theta_1 < \dots < \theta_{R-1}$.

1.1.12.1 Cumulative-Logit Probabilities Define the logistic CDF (argument clamped to $[-30, 30]$) and density:

$$F_k = \text{logistic}(\theta_k - \mu) = \frac{1}{1 + e^{-(\theta_k - \mu)}}, \quad f_k = F_k(1 - F_k)$$

with boundary conventions $F_0 = 0$, $F_R = 1$. Category probabilities:

$$\pi_r = F_r - F_{r-1}, \quad r = 1, \dots, R.$$

Each π_r is floored at $\text{MIN_PROB} = 10^{-10}$ and the vector is renormalized to sum to 1.

1.1.12.2 Log-Likelihood

$$\ell_i = \log \pi_{y_i} = \log(F_{y_i} - F_{y_i-1})$$

1.1.12.3 Derivatives for μ (identity link) Using boundary conventions $f_0 = f_R = 0$, for observation $y_i = r$:

Score:

$$u_\mu = \frac{f_{r-1} - f_r}{\pi_r}$$

Expected Fisher information (sum over all categories):

$$w_\mu = \sum_{r=1}^R \frac{(f_{r-1} - f_r)^2}{\pi_r}$$

Implementation:

$$u_\mu = \frac{f_{r-1} - f_r}{\pi_r}, \quad w_\mu = \sum_{r=1}^R \frac{(f_{r-1} - f_r)^2}{\pi_r}$$

Both floored at MIN_WEIGHT where needed.

1.1.12.4 Derivatives for δ_k (identity link for $k = 1$, log link for $k \geq 2$) The Jacobian of the $\eta_k \rightarrow \theta_k$ map is:

$$\text{jac}_k = \begin{cases} 1 & k = 1 \text{ (identity link)} \\ \exp(\eta_k) = \delta_k & k \geq 2 \text{ (log link; stored as response-scale value)} \end{cases}$$

For observation $y_i = r$, threshold δ_k only affects π_r when $r \geq k$:

Score:

$$u_k = \begin{cases} 0 & r < k \\ \frac{\text{jac}_k (f_r \mathbf{1}\{r \leq R-1\} - f_{r-1} \mathbf{1}\{r > k\})}{\pi_r} & r \geq k \end{cases}$$

Expected Fisher information (sum over categories $r = k, \dots, R$):

$$w_k = \sum_{r=k}^R \frac{\left(\text{jac}_k (f_r \mathbf{1}\{r \leq R-1\} - f_{r-1} \mathbf{1}\{r > k\}) \right)^2}{\pi_r}$$

floored at MIN_WEIGHT.

1.1.12.5 Moments

$$\mathbb{E}[Y] = \sum_{r=1}^R r \pi_r, \quad \text{Var}(Y) = \sum_{r=1}^R r^2 \pi_r - (\mathbb{E}[Y])^2$$

(variance clamped to ≥ 0).

Implementation note: `Ocat` carries `n_categories` state and cannot be recovered from a name string alone. It is excluded from `from_name` and from the WASM / JSON string-dispatch routes. Construct it explicitly in Rust or via the Python `Ocat(n_categories=R)` class. The fitting loop treats each threshold δ_k as an intercept-only formula — no new solver machinery is required beyond what drives any other scalar distribution parameter.

Source: `src/distributions/ocat.rs`.

1.1.13 1.9 Example: Computing Derivatives for Gaussian Data

Suppose we have data $y = [2.1, 1.9, 3.2]$ and current parameter estimates $\mu = [2.0, 2.0, 3.0]$, $\sigma = [0.5, 0.5, 0.5]$.

For μ (identity link):

Residuals: $r = y - \mu = [0.1, -0.1, 0.2]$

Weights: $w_\mu = 1/\sigma^2 = [4.0, 4.0, 4.0]$

Score: $u_\mu = r \cdot w_\mu = [0.4, -0.4, 0.8]$

Working response: $z_\mu = \mu + u_\mu/w_\mu = \mu + r = y$

(For identity link with constant weights, the working response is just y)

For σ (log link):

Squared residuals: $r^2 = [0.01, 0.01, 0.04]$

Score on $\eta_\sigma = \log(\sigma)$ scale:

$$u_\sigma = \frac{r^2 - \sigma^2}{\sigma^2} = \frac{[0.01, 0.01, 0.04] - 0.25}{0.25} = [-0.96, -0.96, -0.84]$$

Weights: $w_\sigma = [2.0, 2.0, 2.0]$

Working response on log scale:

$$z_\sigma = \log(\sigma) + u_\sigma/w_\sigma = [-0.693, -0.693, -0.693] + [-0.48, -0.48, -0.42] = [-1.17, -1.17, -1.11]$$

The PWLS solver then regresses z_σ on the design matrix for σ with weights w_σ to update σ .

1.1.14 1.10 Cumulative Distributions, Densities, and Quantiles

Beyond the score/Fisher quantities that drive fitting, each family exposes the **distributional trio** — the cumulative distribution F , the density/mass f , and the quantile (inverse CDF) Q . These are what every statistically distinctive GAMLSS output needs (quantile residuals in §11.3, centile curves, predictive intervals).

1.1.14.1 Definitions

$$F(y | \theta) = P(Y \leq y | \theta), \quad f(y | \theta) = \begin{cases} \frac{\partial F}{\partial y} & \text{continuous} \\ F(y) - F(y^-) & \text{discrete} \end{cases}, \quad Q(p | \theta) = \inf\{y : F(y | \theta) \geq p\}.$$

The density is obtained for free from the pointwise log-likelihood of §1.1–§1.8:

$$f(y | \theta) = \exp(\ell_{\text{pointwise}}(y | \theta)),$$

which is exact because every family's $\ell_{\text{pointwise}}$ returns a **normalized** log-density (continuous) or log-mass (discrete). Discrete families set an `is_discrete` flag; their F is the right-continuous step function evaluated at $\lfloor y \rfloor$.

1.1.14.2 Why the trio matters: the probability integral transform If Y is continuous with CDF F , then

$$U = F(Y | \theta) \sim \text{Uniform}(0, 1) \implies r = \Phi^{-1}(U) \sim N(0, 1).$$

So $r_i = \Phi^{-1}(F(y_i | \hat{\theta}_i))$ is standard normal whenever the fitted model is correct, **regardless of the family** — the foundation of the randomized quantile residuals of §11.3.

1.1.14.3 Per-family CDF and quantile Each family maps its gamlss (mean/dispersion) parameters to the standard arguments of one special function. All special functions are in **stats**: the error function erf ; the regularized lower / upper incomplete gamma $P(a, x) = \gamma(a, x)/\Gamma(a)$ and $Q(a, x) = 1 - P(a, x)$; and the regularized incomplete beta $I_x(a, b) = B(x; a, b)/B(a, b)$.

Family	gamlss params	standard params	$F(y \mid \theta)$	$Q(p \mid \theta)$
Gaussian	μ, σ	loc μ , sd σ	$\frac{1}{2}[1 + \text{erf}(\frac{y-\mu}{\sigma\sqrt{2}})]$	$\mu + \sigma\sqrt{2} \text{erf}^{-1}(2p - 1)$
Gamma	μ, σ	shape $\alpha = 1/\sigma^2$, scale $s = \mu\sigma^2$	$P(\alpha, y/s)$	$\text{Gamma}(\alpha, 1/s)^{-1}(p)$
Student-t	μ, σ, ν	std t on $z = (y - \mu)/\sigma$	$T_\nu(z)$	$\mu + \sigma T_\nu^{-1}(p)$
Beta	μ, ϕ	$a = \mu\phi$, $b = (1 - \mu)\phi$	$I_y(a, b)$	$\text{Beta}(a, b)^{-1}(p)$
Poisson (<i>disc.</i>)	μ	rate μ	$Q(\lfloor y \rfloor + 1, \mu)$	search (below)
Neg. binomial (<i>disc.</i>)	μ, σ	size $r = 1/\sigma$, prob $q = r/(r + \mu)$	$I_q(r, \lfloor y \rfloor + 1)$	search
Binomial (<i>disc.</i>)	n (state), μ	n trials, prob μ	$I_{1-\mu}(n - \lfloor y \rfloor, \lfloor y \rfloor + 1)$	search
Ocat (<i>disc.</i>)	μ, δ thresholds	cumulative-logit	$\text{logistic}(\theta_{\lfloor y \rfloor} - \mu)$, $= 1$ at $y \geq R$	smallest level r with $F(r) \geq p$

Note Beta uses the **precision** parameterization (μ, ϕ) consistent with §1.6 and the glossary, so $a = \mu\phi$ and $b = (1 - \mu)\phi$ (not a (μ, σ) form).

1.1.14.4 Discrete CDFs via incomplete special functions The discrete CDFs avoid a summation loop through a special-function identity. For the Poisson (representative of the discrete cases),

$$F(m \mid \mu) = \sum_{k=0}^m e^{-\mu} \frac{\mu^k}{k!} = Q(m+1, \mu) = \frac{\Gamma(m+1, \mu)}{\Gamma(m+1)}, \quad m = \lfloor y \rfloor,$$

i.e. the regularized **upper** incomplete gamma at an integer first argument equals the Poisson CDF. The negative-binomial and binomial CDFs are the regularized incomplete beta evaluations in the table; Ocat is the proportional-odds cumulative-logit $P(Y \leq r) = \text{logistic}(\theta_r - \mu)$ (the same F_k used in §1.8).

1.1.14.5 Discrete quantile search For the discrete families, Q is the smallest integer k with $F(k) \geq p$, found by a doubling-bracket then bisection (the shared **discrete_quantile** helper):

double hi until $F(hi) \geq p$, then bisect $[lo, hi]$ to the smallest k with $F(k) \geq p$,

which is $O(\log k)$ even for large means. The right-continuous convention — evaluating the discrete CDF at $\lfloor y \rfloor$ — gives the identity

$$F(k) - F(k-1) = f(k) \quad (\text{pmf at integer support}),$$

which the randomized quantile residuals of §11.3 rely on to de-lump each atom.

1.2 2. Special Functions

1.2.1 2.1 Digamma Function

The digamma function $\psi(x) = \frac{d}{dx} \log \Gamma(x) = \frac{\Gamma'(x)}{\Gamma(x)}$ appears in derivatives for Gamma, Negative Binomial, and Beta distributions.

1.2.1.1 Recurrence Relation For $x < 5$, use:

$$\psi(x) = \psi(x+1) - \frac{1}{x}$$

Repeatedly apply until $x \geq 5$.

1.2.1.2 Asymptotic Expansion For $x \geq 5$, use the series (Abramowitz & Stegun 6.3.18):

$$\psi(x) \sim \log(x) - \frac{1}{2x} - \frac{1}{12x^2} + \frac{1}{120x^4} - \frac{1}{252x^6} + O(x^{-8})$$

1.2.1.3 Known Values (for testing)

- $\psi(1) = -\gamma \approx -0.5772156649$ (Euler-Mascheroni constant)
- $\psi(2) = 1 - \gamma \approx 0.4227843351$
- $\psi(1/2) = -\gamma - 2\log(2) \approx -1.9635100260$

1.2.2 2.2 Trigamma Function

The trigamma function $\psi'(x) = \frac{d^2}{dx^2} \log \Gamma(x)$ is needed for Student-t Fisher information.

1.2.3 Recurrence Relation

For $x < 5$, use:

$$\psi'(x) = \psi'(x+1) + \frac{1}{x^2}$$

1.2.4 Asymptotic Expansion

For $x \geq 5$, use the series (Abramowitz & Stegun 6.4.11):

$$\psi'(x) \sim \frac{1}{x} + \frac{1}{2x^2} + \frac{1}{6x^3} - \frac{1}{30x^5} + \frac{1}{42x^7} - \frac{1}{30x^9} + O(x^{-11})$$

1.2.5 Known Values (for testing)

- $\psi'(1) = \frac{\pi^2}{6} \approx 1.644934067$
 - $\psi'(2) = \frac{\pi^2}{6} - 1 \approx 0.644934067$
 - $\psi'(1/2) = \frac{\pi^2}{2} \approx 4.934802201$
-

1.3 3. Numerical Stability and Implementation

1.3.1 3.1 Parameter Bounds

To prevent numerical issues, parameters are clamped to safe ranges:

Minimum positive value: 10^{-10} - Used for: μ (when must be positive), σ , ϕ , probabilities

Minimum weight: 10^{-6} - Fisher information weights are floored at this value to ensure positive definiteness of the weight matrix

Link function bounds: - Log/logit links: $\eta \in [-30, 30]$ - Prevents overflow in $\exp(\eta)$ (since $e^{30} \approx 10^{13}$ and $e^{-30} \approx 10^{-14}$)

1.3.2 3.2 Distribution-Specific Safeguards

Student-t: - Enforce $\nu > 2$ (finite variance) via the floored log link on ν (§7) - Ensure $\nu + z^2 \geq 10^{-10}$ to prevent division issues in robustifying weight

Gamma: - Clamp σ to reasonable range to prevent extreme shape parameters

Beta: - Clamp y to $(10^{-10}, 1 - 10^{-10})$ before computing $\log(y)$ and $\log(1 - y)$ - Clamp μ similarly before computing logit

Negative Binomial: - Ensure $1 + \sigma\mu$ stays positive and bounded away from zero

Ordered Categorical: - Floor each category probability π_r at $\text{MIN_PROB} = 10^{-10}$ before computing scores and Fisher info - Renormalize: divide all π_r by their sum after flooring, so that $\sum_r \pi_r = 1$ is maintained

1.3.3 3.3 Batched Special Function Computation

Digamma and trigamma functions are computed in batched vectorized form for performance:

- For $n < 10,000$ observations: sequential computation
- For $n \geq 10,000$: parallel computation via Rayon (when `parallel` feature enabled)

The switchpoint is empirically tuned to balance overhead vs. speedup.

1.4 4. GAMLSS Iteration (P-IRLS)

1.4.1 4.1 Backfitting Outer Loop

GAMLSS fits the joint model by cycling through distribution parameters one at a time, holding the others fixed (Rigby & Stasinopoulos 2005, §3). For a P -parameter family (e.g. Gaussian: $P = 2$; Student-t: $P = 3$):

```
for cycle in 0..max_iterations:
    all_converged = true
    for each parameter theta_k in family.parameters():
        eta_k_old = eta_k
        eta_k      = p_irls_step(theta_k | other params held fixed)    # eta = X * beta
        delta      = max_abs(eta_k - eta_k_old)
        scale      = max(max_abs(eta_k), 1.0)    # floor at 1 for 0(1) predictors
        if delta / scale >= tolerance:
            all_converged = false
    gd = global_deviance(all params)    # -2 * sum log f(y_i | theta_i)
    if all_converged and abs(gd_prev - gd) < gd_tolerance:
        break
    gd_prev = gd
```

Source: `src/fitting/mod.rs`. Convergence requires **two tests to pass together**:

1. **Per-parameter relative change of the linear predictor** $\eta_k = X_k \beta_k$:

$$\frac{\|\eta_k^{(t+1)} - \eta_k^{(t)}\|_\infty}{\max(\|\eta_k^{(t+1)}\|_\infty, 1)} < \epsilon$$

for every k . The test is in **fit space** rather than coefficient space: penalized designs can carry fit-irrelevant coefficient ridges (a flat REML valley where the per-cycle λ re-optimization jitters between fit-equivalent (λ, β) pairs) along which β never becomes stationary even though the model — η , μ , the deviance — already is. Using each parameter's own scale prevents a large parameter (e.g. μ on

un-normalized data) from masking drift in a small one (e.g. a log-scale σ near 0); the floor of 1 keeps the test equivalent to an absolute threshold for $O(1)$ predictors.

2. **Absolute global-deviance change** $|GD^{(t)} - GD^{(t+1)}| < \epsilon_{GD}$, the same convention as R gamlss's `c.crit` (an earlier *relative* form scaled its slack with $|GD|$ and stopped large-deviance fits far short of the optimum).

The active defaults are $\epsilon = 10^{-3}$, $\epsilon_{GD} = 10^{-3}$, max iterations = 200 (DEFAULT_TOLERANCE, DEFAULT_GD_TOLERANCE, DEFAULT_MAX_ITER in `src/fitting/mod.rs`).

1.4.2 4.2 Inner P-IRLS Step (Fisher Scoring)

For each parameter θ_k (e.g., μ , σ , ν):

1. **Working response:**

$$z_k = \eta_k + \frac{u_k}{w_k}$$

where $u_k = \frac{\partial \ell}{\partial \eta_k}$ and w_k is the (expected, or in some cases observed) Fisher information.

2. **Penalized weighted least squares** (see §8.1 for the Cholesky solve):

$$\hat{\beta}_k = \arg \min_{\beta} \|W_k^{1/2}(z_k - X_k \beta)\|^2 + \sum_j \lambda_j \beta^T S_j \beta$$

3. **Update linear predictor:**

$$\eta_k^{(\text{new})} = X_k \hat{\beta}_k$$

4. **Smoothing parameter selection** — GCV, REML, or Fellner–Schall (§§ 8, 8.2).

5. **Step-halving on the penalized deviance** (FIT-1): the accepted update is $\beta_k^{(t)} + \alpha d_k$ with $d_k = \hat{\beta}_k - \beta_k^{(t)}$ and $\alpha \in \{1, \frac{1}{2}, \frac{1}{4}, \dots\}$ backtracked until

$$GD(\beta_k^{(t)} + \alpha d_k) + \sum_j \lambda_j (\beta_k^{(t)} + \alpha d_k)^T S_j (\beta_k^{(t)} + \alpha d_k)$$

does not increase. The objective must be the **penalized** deviance: the Fisher direction d_k is an ascent direction for the penalized log-likelihood only, and when the current β is wigglier than the penalized optimum (e.g. λ grew across cycles) a raw-deviance line search rejects every step and freezes the fit at a non-stationary point. The penalty change along the path is evaluated in the cancellation-free form $2\alpha d^T S_\lambda \beta + \alpha^2 d^T S_\lambda d$ so that a clamp-ceiling λ ($\sim e^{30}$) cannot swamp the comparison with round-off. If the backtracking floor ($\alpha = 2^{-10}$) is reached with the objective still increasing, the block update is **rejected** ($\alpha = 0$) rather than forced — accepting uphill micro-steps allows unbounded slow divergence.

Source: `src/fitting/scoring.rs`.

1.4.3 4.3 Working-Response Derivation

Why is $z = \eta + u/w$ the right pseudo-response? Expand the log-likelihood for one observation in η around the current iterate η_0 :

$$\ell(\eta) \approx \ell(\eta_0) + u(\eta_0)(\eta - \eta_0) - \frac{1}{2} w(\eta_0)(\eta - \eta_0)^2$$

where $u = \partial \ell / \partial \eta$ and $w = -\mathbb{E}[\partial^2 \ell / \partial \eta^2]$ (Fisher information, positive). Maximizing the quadratic Taylor surrogate over η gives the one-Newton-step update $\eta - \eta_0 = u/w$. Setting $\eta = X\beta$ and stacking observations yields the Gauss–Newton normal equations:

$$X^T W X \Delta\beta = X^T W (u/w)$$

with $\Delta\beta = \beta - \beta_0$ and the diagonal weight $W = \text{diag}(w_i)$. Equivalently, regress the **working response** $z = \eta_0 + u/w$ on X with weights W :

$$X^T W X \beta = X^T W z.$$

Adding the penalty $\sum_j \lambda_j \beta^T S_j \beta$ to the surrogate produces the PWLS system of §8.1. This is exactly the IRLS construction of McCullagh & Nelder (1989) generalized to one distributional parameter at a time.

1.4.4 4.4 Numerical Safeguards in the Inner Loop

Cataloged here so the magic constants are auditable. All apply per observation, per inner iteration. Source: `src/fitting/scoring.rs`, `src/distributions/links.rs`.

Symbol	Value	Where	Purpose
MIN_WEIGHT	10^{-6}	<code>scoring.rs</code>	Floor on w_i so the weight matrix stays positive definite; near-zero Fisher info would blow up u/w . Hits counted in <code>weight_floor_hits</code> .
MAX_STEP	10^6	<code>scoring.rs</code>	Clamps u_i/w_i in η -units — a pure anti-overflow guard for degenerate score/information combinations. It must be large : a tight clamp (an earlier value of 20) silently biased the working response, because when many rows clip, the update direction is decided by the <i>count</i> of positive vs negative rows rather than the score-weighted aggregate — which can point the Fisher step uphill. Overshoot robustness is provided by the deviance-guarded step-halving instead. Hits counted in <code>step_cap_hits</code> .
MAX_ETA, MIN_ETA	± 30	<code>links.rs</code>	Clamps η before applying the inverse log/logit link so $\exp(\eta)$ stays finite ($e^{30} \approx 10^{13}$).

Symbol	Value	Where	Purpose
MIN_POSITIVE	10^{-10}	links.rs, diagnostics.rs	Floor on parameters that must be strictly positive (σ , ϕ , probabilities), and on variances before the square-root in residual computation.

Persistent non-zero `weight_floor_hits` or `step_cap_hits` at convergence signals model misspecification or ill-conditioned data; transient hits in early iterations are normal.

1.4.5 4.5 Prior / Observation Weights

Each call to the P-IRLS `step` function accepts an optional `prior_weights: Option<Array1<f64>>` — a per-observation scale applied to each observation’s likelihood contribution. When absent, a ones vector is substituted (uniform weights).

Let $\text{safe_w}_i = \max(w_i, \text{MIN_WEIGHT})$ where w_i is the distribution’s Fisher information for observation i .

Working response — unchanged by prior weights:

$$z_i = \eta_i + \text{clip}\left(\frac{u_i}{\text{safe_w}_i}, \pm \text{MAX_STEP}\right)$$

The step size $u_i/\text{safe_w}_i$ is a property of log-likelihood curvature and must not be scaled by the observation weight.

Solve weight — carries the prior factor:

$$W_{ii} = \text{prior}_i \cdot \text{safe_w}_i$$

Effect on the penalized normal equations: because z divides by safe_w while W multiplies by $\text{prior} \cdot \text{safe_w}$, the prior weight scales each observation’s contribution to both $X^T W X$ and $X^T W z$ exactly once — standard frequency / precision-weight semantics without distorting the IRLS step length:

$$(X^T W X + S_\lambda) \hat{\beta} = X^T W z, \quad W = \text{diag}(\text{prior}_i \cdot \text{safe_w}_i).$$

When `prior_weights = None`, the formula reduces to $W = \text{diag}(\text{safe_w}_i)$, identical to the unweighted case.

Source: `src/fitting/scoring.rs (step)`.

1.5 5. Smoothing and Penalty Matrices

1.5.1 5.0 B-Spline Basis (Cox–de Boor)

A degree- p B-spline basis with k functions is defined on a knot vector

$$\tau_0 < \tau_1 < \dots < \tau_{k+p}.$$

Knot placement (P-spline, `bs="ps"`). The implementation uses **equally-spaced** knots — the requirement of the Eilers–Marx P-spline, because the difference penalty $D^T D$ only approximates a roughness integral when the basis knots are uniform:

$$\tau_j = \min(x) + (j - p) \Delta, \quad \Delta = \frac{\max(x) - \min(x)}{k - p}, \quad j = 0, \dots, k + p.$$

This places $\tau_p = \min(x)$ and $\tau_k = \max(x)$, with p extra knots extending beyond each end of the data range. The anchoring range $(\min x, \max x)$ is resolved **once from the training data** and stored on the term (`PSpline1D::range`, `TensorProduct::range_1/2`), so prediction on a grid, a subset, or out-of-range points is evaluated on the *training* knot grid — re-deriving the range from the prediction data (the previous behavior) silently applied the coefficients to a different basis. Source: `src/splines/pspline.rs` (`select_knots`, `create_basis_matrix_with_range`), `src/fitting/assembler.rs` (`resolve_terms`).

The **Cox–de Boor recursion** evaluates the basis stably (Piegl & Tiller 1997, Algorithm A2.2). Let i be the **knot span** — the index satisfying $\tau_i \leq x < \tau_{i+1}$, clamped to $[p, k-1]$. Initialize $N_0 = 1$. Then for $j = 1, \dots, p$:

$$\text{left}[j] = x - \tau_{i+1-j}, \quad \text{right}[j] = \tau_{i+j} - x,$$

and for $r = 0, \dots, j-1$:

$$\begin{aligned} \text{denom} &= \text{right}[r+1] + \text{left}[j-r], & \text{term} &= N_r / \text{denom}, \\ N_r &\leftarrow N_r^{\text{saved}} + \text{right}[r+1] \cdot \text{term}, & N_r^{\text{saved}} &\leftarrow \text{left}[j-r] \cdot \text{term}. \end{aligned}$$

At the end of iteration j , $N_0, \dots, N_j$ hold the $j+1$ non-zero degree- j basis values at x . **Critical implementation note:** `left[j]` and `right[j]` must be computed *once, before the inner r -loop*, so that the full arrays `left[1..j]` and `right[1..j]` remain valid when the inner loop reads `left[j-r]` and `right[r+1]`. Computing them inside the r -loop overwrites earlier slots and produces wrong interior basis values for degree ≥ 3 (the endpoint functions are unaffected but the middle ones are wrong; both the correct and the broken form still satisfy partition-of-unity, so property-only tests do not catch the error).

After the full degree- p pass, $N_0, \dots, N_p$ are the $p+1$ non-zero values at x ; they map to columns $[i-p, i]$ of the basis matrix. Source: `src/splines/pspline.rs` (`create_basis_matrix_with_range`, `evaluate_basis_functions_into`, `find_knot_span`).

Properties:

- **Partition of unity:** $\sum_{j=1}^k B_j(x_i) = 1$ for every x_i in the interior of the knot range.
- **Non-negativity:** $B_j(x) \geq 0$.
- **Local support:** $B_{i,p}$ is non-zero only on $[\tau_i, \tau_{i+p+1}]$, so each row of the design matrix has at most $p+1$ non-zero entries.

1.5.2 5.1 Sum-to-Zero Reparameterization (Identifiability)

Partition-of-unity creates a rank deficiency when the model has both an intercept and a smooth term: the column vector $\mathbf{1}_n$ lies in the column space of B , so the design matrix $[\mathbf{1}_n \mid B]$ is column-rank-deficient. To restore identifiability, the smooth is reparameterized through an orthonormal basis of the subspace orthogonal to $\mathbf{1}_k$.

The implementation uses a **Householder reflector** (`src/splines/`). Choose

$$v = \mathbf{1}_k + \sqrt{k} e_1$$

which reflects $\mathbf{1}_k$ onto $-\sqrt{k} e_1$. Form

$$H = I_k - \frac{2}{\|v\|^2} vv^T, \quad Z = H_{:,2:k}$$

i.e. drop the first column of H . Then $Z \in \mathbb{R}^{k \times (k-1)}$ satisfies

$$Z^T Z = I_{k-1}, \quad Z^T \mathbf{1}_k = 0.$$

Reparameterize $\beta = Z\gamma$ to get the constrained basis and penalty:

$$B_{\text{new}} = BZ, \quad S_{\text{new}} = Z^T S Z.$$

The new design matrix $[\mathbf{1}_n \mid BZ]$ has full column rank, and the constraint $\mathbf{1}_k^T \beta = 0$ — i.e. the smooth has mean zero across knots — is enforced automatically.

1.5.3 P-Spline Penalty (Eilers & Marx 1996)

For 2nd-order differences:

$$S = D^T D$$

where D is the $(m-2) \times m$ difference matrix:

$$D = \begin{bmatrix} 1 & -2 & 1 & 0 & \cdots & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 \\ \vdots & & \ddots & & & \vdots \\ 0 & \cdots & 0 & 1 & -2 & 1 \end{bmatrix}$$

This penalizes curvature: $\lambda \beta^T S \beta \approx \lambda \int (\beta''(x))^2 dx$

1.5.4 Tensor Product Smooth and Anisotropic Penalty

For a bivariate smooth $f(x_1, x_2)$ built from marginal bases $B_1 \in \mathbb{R}^{n \times k_1}$ and $B_2 \in \mathbb{R}^{n \times k_2}$, the joint basis is the **row-wise Kronecker product** of the marginals:

$$B[i, :] = B_1[i, :] \otimes B_2[i, :] \in \mathbb{R}^{k_1 k_2}, \quad B \in \mathbb{R}^{n \times (k_1 k_2)}.$$

The coefficient vector $\beta \in \mathbb{R}^{k_1 k_2}$ is indexed lexicographically so $\beta_{(i-1)k_2+j}$ multiplies $B_1[\cdot, i] \cdot B_2[\cdot, j]$.

Anisotropic penalties are obtained by Kronecker'ing each marginal difference penalty with an identity matrix:

$$S^{(1)} = S_1 \otimes I_{k_2}, \quad S^{(2)} = I_{k_1} \otimes S_2$$

with two independent smoothing parameters:

$$\lambda_1 \beta^T S^{(1)} \beta + \lambda_2 \beta^T S^{(2)} \beta.$$

$S^{(1)}$ penalizes roughness in the x_1 direction at every x_2 slice; $S^{(2)}$ vice versa. Choosing two λ 's lets the GCV/REML routine pick different smoothness for each margin (Wood 2017, §5.6).

When an intercept is present, **one** sum-to-zero constraint (§5.1) is applied to the **full** tensor basis, transforming both penalties with the same $Z \in \mathbb{R}^{k_1 k_2 \times (k_1 k_2 - 1)}$:

$$B_{\text{new}} = BZ, \quad S_{\text{new}}^{(j)} = Z^T S^{(j)} Z, \quad j = 1, 2.$$

This is mgcv's `te()` treatment ($k_1 k_2 - 1$ coefficients) and removes exactly the constant function — the only direction collinear with the intercept, since the row-Kronecker of two partition-of-unity bases is itself partition-of-unity ($B \mathbf{1}_{k_1 k_2} = \mathbf{1}_n$).

Centering each **margin** before the Kronecker product ($B_1 Z_1 \otimes_{\text{row}} B_2 Z_2$) is *not* equivalent: the product of two constant-free marginal spaces excludes every function of the form $f(x_1) \cdot 1$ and $1 \cdot g(x_2)$ — both main effects — leaving a pure-interaction (`ti()`-style) smooth that cannot represent additive structure. An earlier implementation made exactly this error. Consequence of the correct construction: the penalty null space after centering has dimension $d_1 d_2 - 1$ (e.g. 3 for order-2 margins, matching mgcv), and a formula combining `te(x1, x2)` with a separate main effect of x_1 on the same parameter is rank-deficient — use a pure-interaction decomposition for that, as in mgcv.

Source: `src/fitting/assembler.rs` (`assemble_smooth`), `src/splines/reparam.rs`.

1.5.5 Random Effect Penalty

For an observation-to-group map $g : \{1, \dots, n\} \rightarrow \{1, \dots, G\}$, the design matrix is the binary indicator

$$Z \in \{0, 1\}^{n \times G}, \quad Z[i, g] = \mathbf{1}\{g(i) = g\}.$$

Coupled with the **identity penalty** $S = I_G$, the model

$$\eta = \mathbf{1}\beta_0 + Z\alpha, \quad \lambda\alpha^T\alpha = \lambda \sum_{g=1}^G \alpha_g^2$$

is equivalent to the Bayesian random-intercept prior $\alpha_g \sim N(0, 1/\lambda)$ — the ridge-penalized empirical Bayes interpretation. Since each row of Z sums to 1, the sum-to-zero reparameterization of §5.1 is again applied when an intercept is present:

$$Z_{\text{new}} = Z Z_c \in \mathbb{R}^{n \times (G-1)}, \quad Z_c^T I_G Z_c = I_{G-1}.$$

The group-to-column map is resolved **once at fit time** (levels sorted for determinism) and stored on the term, so prediction maps each group to the coefficient it was fitted with regardless of the row order or group subset of the new data; an unseen level at predict time is an error (mgcv factor semantics). Source: `src/fitting/assembler.rs` (`resolve_terms`, `assemble_smooth`).

1.5.6 5.1 Block-Sparse Penalty Matrices

In practice, penalty matrices S_j are often **block-sparse**: they only penalize a subset of the coefficient vector. For example, if the model has an intercept, linear terms, and a smooth term:

$$\beta = [\beta_{\text{intercept}}, \beta_{\text{linear}}, \beta_{\text{smooth}}]^T$$

The penalty for the smooth term is:

$$S_{\text{smooth}} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & S_{\text{block}} \end{bmatrix}$$

where S_{block} is the $(n_{\text{splines}} \times n_{\text{splines}})$ penalty for the spline coefficients.

Efficient Storage: Instead of storing the full $p \times p$ matrix (mostly zeros), we store: - **block**: The non-zero sub-matrix S_{block} - **offset**: Starting index in the full coefficient vector - **full_dim**: Total dimension p

Operations:

Scaled addition: $A \leftarrow A + \lambda S$

$$A[i : i + k, j : j + k] \leftarrow A[i : i + k, j : j + k] + \lambda \cdot S_{\text{block}}$$

where $i = \text{offset}$ and $k = \text{dim}(S_{\text{block}})$.

Matrix-vector product: $v = S\beta$

$$v[i : i + k] = S_{\text{block}} \cdot \beta[i : i + k], \quad v[\text{elsewhere}] = 0$$

This exploits sparsity for computational efficiency, especially important when many terms have independent penalties.

1.6 6. Effective Degrees of Freedom

The “wiggleness” of the fit is measured by:

$$\text{EDF} = \text{tr}(H)$$

where $H = X(X^T W X + \lambda S)^{-1} X^T W X$ is the hat matrix.

Interpretation: - EDF ≈ 2 for a linear fit (intercept + slope) - EDF $\approx$ number of basis functions for an unpenalized spline - EDF between these extremes reflects smoothing

1.7 7. Link Functions

1.7.1 Identity Link

- **Function:** $g(\mu) = \mu$
- **Inverse:** $g^{-1}(\eta) = \eta$
- **Domain:** $\mu \in \mathbb{R}$
- **Use:** Gaussian mean, Student-t location
- **Derivative:** $\frac{d\mu}{d\eta} = 1$

1.7.2 Log Link

- **Function:** $g(\mu) = \log(\mu)$
- **Inverse:** $g^{-1}(\eta) = \exp(\eta)$
- **Domain:** $\mu > 0$
- **Use:** Poisson rate, Gamma mean, scale parameters (σ)
- **Derivative:** $\frac{d\mu}{d\eta} = \mu$
- **Numerical safeguard:** Clamp $\eta \in [-30, 30]$ to prevent overflow

1.7.3 Floored Log Link

- **Function:** $g(\mu) = \log(\max(\mu, c))$
- **Inverse:** $g^{-1}(\eta) = \max(\exp(\eta), c)$
- **Domain:** $\mu \geq c$
- **Use:** Student-t degrees of freedom ν with floor $c = 2$, keeping the variance $\sigma^2\nu/(\nu - 2)$ finite as the optimizer explores the heavy-tail region
- **Derivative:** $\frac{d\mu}{d\eta} = \mu$ above the floor (the floor binds only transiently and never when the true ν is well above 2)
- **Numerical safeguard:** same $\eta \in [-30, 30]$ clamp as the log link, plus the lower floor c . See `FlooredLogLink` in `src/distributions/links.rs`.

1.7.4 Logit Link

- **Function:** $g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$
- **Inverse:** $g^{-1}(\eta) = \frac{1}{1+e^{-\eta}} = \frac{e^{\eta}}{1+e^{\eta}}$
- **Domain:** $\mu \in (0, 1)$
- **Use:** Binomial probability, Beta mean
- **Derivative:** $\frac{d\mu}{d\eta} = \mu(1 - \mu)$
- **Numerical safeguards:**
 - Clamp $\mu \in [10^{-10}, 1 - 10^{-10}]$ before applying link
 - Clamp $\eta \in [-30, 30]$ to prevent overflow in inverse

1.7.5 Chain Rule for Link Functions

When modeling parameter θ with link function g , we have $\eta = g(\theta)$. The score and Fisher information transform as:

$$\frac{\partial \ell}{\partial \eta} = \frac{d\theta}{d\eta} \cdot \frac{\partial \ell}{\partial \theta}$$

$$I_\eta = \left(\frac{d\theta}{d\eta} \right)^2 I_\theta$$

For log link ($\eta = \log(\theta)$):

$$\frac{d\theta}{d\eta} = \theta, \quad \frac{\partial \ell}{\partial \eta} = \theta \cdot \frac{\partial \ell}{\partial \theta}, \quad I_\eta = \theta^2 I_\theta$$

For logit link ($\eta = \text{logit}(\theta)$):

$$\frac{d\theta}{d\eta} = \theta(1 - \theta), \quad \frac{\partial \ell}{\partial \eta} = \theta(1 - \theta) \cdot \frac{\partial \ell}{\partial \theta}, \quad I_\eta = [\theta(1 - \theta)]^2 I_\theta$$

For identity link ($\eta = \theta$):

$$\frac{d\theta}{d\eta} = 1, \quad \frac{\partial \ell}{\partial \eta} = \frac{\partial \ell}{\partial \theta}, \quad I_\eta = I_\theta$$

1.7.6 Working Response and IRLS

The working response in the IRLS algorithm is:

$$z = \eta + \frac{u}{w}$$

where $u = \frac{\partial \ell}{\partial \eta}$ and $w = I_\eta$.

For a Poisson model with log link: - Current: $\eta = \log(\hat{\mu})$ - Score: $u = y - \hat{\mu}$ - Weight: $w = \hat{\mu}$ - Working response: $z = \log(\hat{\mu}) + \frac{y - \hat{\mu}}{\hat{\mu}}$

For a Binomial model with logit link: - Current: $\eta = \text{logit}(\hat{\mu})$ - Score: $u = y - n\hat{\mu}$ - Weight: $w = n\hat{\mu}(1 - \hat{\mu})$ - Working response: $z = \text{logit}(\hat{\mu}) + \frac{y - n\hat{\mu}}{n\hat{\mu}(1 - \hat{\mu})}$

The PWLS problem then becomes:

$$\hat{\beta}^{(t+1)} = \arg \min_{\beta} \sum_i w_i (z_i - x_i^T \beta)^2 + \sum_j \lambda_j \beta^T S_j \beta$$

1.8 8. GCV Gradient for Smoothing Parameter Optimization

The GCV score is minimized using L-BFGS optimization. The gradient with respect to $\log(\lambda_j)$ requires careful derivation.

1.8.1 Setup

Let $V = (X^T W X + \sum_j \lambda_j S_j)^{-1}$ and $\hat{\beta}(\lambda) = V X^T W z$.

1.8.2 Key Derivatives

Derivative of β with respect to λ_j :

$$\frac{\partial \hat{\beta}}{\partial \lambda_j} = -VS_j\hat{\beta}$$

Derivative of RSS:

$$\frac{\partial \text{RSS}}{\partial \lambda_j} = 2(X^TWr)^T VS_j\hat{\beta}$$

where $r = z - X\hat{\beta}$ are the residuals.

Derivative of EDF:

$$\frac{\partial \text{EDF}}{\partial \lambda_j} = -\text{tr}(VS_jVX^TWX)$$

1.8.3 Full GCV Gradient

Using the quotient rule on $\text{GCV} = \frac{n \cdot \text{RSS}}{(n - \text{EDF})^2}$:

$$\frac{\partial \text{GCV}}{\partial \lambda_j} = \frac{n}{(n - \text{EDF})^3} \left[\frac{\partial \text{RSS}}{\partial \lambda_j} (n - \text{EDF}) + 2 \cdot \text{RSS} \cdot \frac{\partial \text{EDF}}{\partial \lambda_j} \right]$$

1.8.4 Chain Rule for Log-Space

Since we optimize in log-space:

$$\frac{\partial \text{GCV}}{\partial \log(\lambda_j)} = \lambda_j \cdot \frac{\partial \text{GCV}}{\partial \lambda_j}$$

This ensures $\lambda_j > 0$ without explicit constraints.

1.8.5 8.1 Efficient PWLS Solution via Cholesky Decomposition

The penalized weighted least squares problem:

$$\hat{\beta} = \arg \min_{\beta} \|W^{1/2}(z - X\beta)\|^2 + \sum_j \lambda_j \beta^T S_j \beta$$

has the normal equations:

$$(X^T W X + S_{\lambda}) \hat{\beta} = X^T W z$$

where $S_{\lambda} = \sum_j \lambda_j S_j$.

Key observation: $X^T W X + S_{\lambda}$ is symmetric positive definite (SPD) when S_j are positive semi-definite and W has positive diagonal entries.

1.8.5.1 Cholesky Approach

1. Form the weighted design matrix: $X_w = W^{1/2} X$
2. Form the coefficient matrix: $A = X_w^T X_w + S_{\lambda}$
3. Compute Cholesky decomposition: $A = LL^T$ where L is lower triangular
4. Solve $Ly = X^T W z$ for y (forward substitution)
5. Solve $L^T \hat{\beta} = y$ for $\hat{\beta}$ (backward substitution)

Advantages over LU decomposition: - Exploits SPD structure: $O(p^3/3)$ instead of $O(2p^3/3)$ - Numerically stable for well-conditioned systems - Automatic failure detection: Cholesky fails if matrix is not positive definite

Fallback: If Cholesky fails (due to numerical issues or near-singularity), fall back to LU decomposition with partial pivoting.

Robust log-determinant (`log_det_robust`, `src/linalg.rs`): For the REML cost term $\log|X^T W X + S_\lambda|$, the Cholesky log-det is tried first; on failure (evenly-spaced B-spline design matrices can develop tiny negative floating-point pivots) the symmetric eigensolver `dsyev` is used, clamping eigenvalues to 10^{-300} before summing the logs. This means the REML objective gracefully returns a very negative log-det rather than an error, steering the L-BFGS optimizer away from degenerate λ regions.

1.8.5.2 Covariance Matrix The covariance matrix for $\hat{\beta}$ is:

$$\text{Cov}(\hat{\beta}) = (X^T W X + S_\lambda)^{-1} = A^{-1}$$

With Cholesky $A = LL^T$:

$$A^{-1} = (L^T)^{-1} L^{-1}$$

We compute L^{-1} via triangular inversion, then form $(L^{-1})^T L^{-1}$.

1.8.5.3 Gradient Computation for GCV For GCV optimization, we need $\frac{\partial \text{EDF}}{\partial \lambda_j}$:

$$\frac{\partial \text{EDF}}{\partial \lambda_j} = -\text{tr}(V S_j V X^T W X)$$

where $V = A^{-1}$.

Exploiting block structure: When S_j is block-sparse, only compute:

$$\text{tr}(V[i : i + k, :] S_{\text{block}} V[:, i : i + k] X^T W X)$$

This avoids forming the full $p \times p$ products when $S_{\text{block}} \ll p$.

1.8.6 8.2 REML / Laplace-Approximate Marginal Likelihood

GCV is one option for picking the smoothing parameters; the default `criterion` in `FitConfig` is `Reml` (see `src/fitting/mod.rs`). REML in this setting is the Laplace-approximate marginal likelihood (LAML) of Wood (2011), evaluated at the working PWLS step. The selector is exposed as a three-variant enum:

```
pub enum SmoothingCriterion { Gcv, Reml /* default */, FellnerSchall }
```

`Gcv` minimizes the score of §8 via L-BFGS on $\log \lambda$. `Reml` minimizes $-V_r$ (below) via L-BFGS on $\log \lambda$. `FellnerSchall` targets the same $-V_r$ via a deterministic multiplicative fixed-point (§8.3).

1.8.6.1 The LAML objective Let $S_\lambda = \sum_j \lambda_j S_j$. The working-model LAML negative log-marginal likelihood is

$$-V_r(\lambda) = \frac{1}{2} \log|X^T W X + S_\lambda| - \frac{1}{2} \log|S_\lambda|_+ + \text{Gaussian PWLS terms (data, } \sigma)$$

where $|\cdot|_+$ denotes the **pseudo-determinant** — the product of strictly positive eigenvalues — required because S_λ is rank-deficient (its null space contains the unpenalized polynomial directions: constants for order-1 penalties, lines for order-2, etc., plus the unpenalized intercept and linear columns).

$\log |X^T W X + S_\lambda|$ is computed via `log_det_robust` (see §8.1): Cholesky on the fast path; symmetric eigensolver fallback for near-PD matrices with tiny negative floating-point pivots (common for evenly-spaced B-spline designs), clamping eigenvalues to 10^{-300} before the log so the REML optimizer naturally avoids degenerate λ regions instead of crashing.

Source: `src/fitting/solver.rs` (`RemlCost`); `src/linalg.rs` (`log_det_robust`).

1.8.6.2 Pseudo-determinant via grouped block eigendecomposition `penalty_eigen` takes the full list of penalty matrices and their λ values and processes them in **coefficient-block groups**: `penalty_nonzero_block_range` identifies the non-zero row/column range of each penalty, and all penalties that share the exact same range are combined into one group before eigendecomposition.

For each group g :

1. Form the combined scaled block: $B_g = \sum_{j \in g} \lambda_j S_j|_{\text{block}}$
2. Symmetric eigendecompose: $B_g = Q_g \text{diag}(d_1^{(g)}, \dots) Q_g^T$
3. Per-group relative threshold: $\tau_g = \max(\varepsilon \cdot d_{\max}^{(g)}, 10^{-300})$, where $\varepsilon = \text{REML_RANK_TOL_EPS} = 10^{-8}$

Then:

$$\log |S_\lambda|_+ = \sum_g \sum_{d_i^{(g)} > \tau_g} \log d_i^{(g)}, \quad S_\lambda^+ = \text{block-diag}\{B_g^+\}, \quad M_p = \sum_g \text{null_dim}(B_g).$$

Why grouped decomposition? A single global threshold $\tau = \varepsilon \cdot \max(d_{\max}, 1)$ is dominated by the largest- λ term when smoothing parameters span many orders of magnitude (e.g. $\lambda_1 \approx 10^{-8}$ and $\lambda_4 \approx 5 \times 10^{10}$). This misclassifies small- λ non-null directions as null space, flipping the REML gradient sign. The per-group threshold is relative to each block's own spectral norm, eliminating this pollution. The 10^{-300} hard floor prevents collapse when $d_{\max}^{(g)} \approx 0$ (very small λ).

Tensor-product smooths: the two marginal penalties $\lambda_1(S_1 \otimes I_{k_2})$ and $\lambda_2(I_{k_1} \otimes S_2)$ share the same $k_1 k_2$ coefficient block and are therefore combined before eigendecomposition, so the identity $(\lambda_1 S_1 + \lambda_2 S_2)^+ \neq (\lambda_1 S_1)^+ + (\lambda_2 S_2)^+$ is never violated. For disjoint blocks (one smooth per coefficient block) the formula reduces to a per-penalty decomposition.

Source: `src/fitting/solver.rs` (`penalty_eigen`, `penalty_nonzero_block_range`).

1.8.6.3 REML gradient From Wood (2011, §3) the gradient with respect to $\log \lambda_j$ is

$$\frac{\partial(-V_r)}{\partial \log \lambda_j} = \frac{1}{2} \lambda_j \hat{\beta}^T S_j \hat{\beta} - \frac{1}{2} \lambda_j \text{tr}(S_\lambda^+ S_j) + \frac{1}{2} \lambda_j \text{tr}(V S_j)$$

with $V = (X^T W X + S_\lambda)^{-1}$ (working scale $\varphi = 1$; the noise scale enters through W). The implementation evaluates this analytically, validates it against central finite differences in `reml_tests`, and feeds it to L-BFGS; final $\log \lambda$ values are clamped to $[-\text{LOG_LAMBDA_CLAMP}, \text{LOG_LAMBDA_CLAMP}] = [-30, 30]$.

Fellner–Schall polish. L-BFGS with a Moré–Thuente line search can stall at a warm-start-dependent, non-stationary point when the LAML surface has flat ridges (several smooths collapsing onto their null space with λ at the clamp ceiling). The resulting per-cycle λ jitter prevents the outer RS loop from ever seeing a stationary η . `run_optimization_reml` therefore polishes the L-BFGS output with the deterministic Fellner–Schall fixed point (§8.3) on the same target and keeps whichever λ scores better, so the polish can never worsen the fit; a linear-algebra failure inside the polish falls back to the L-BFGS result.

Source: `src/fitting/solver.rs`.

1.8.7 8.3 Fellner–Schall Multiplicative Fixed Point

Wood & Fasiolo (2017) show that for the LAML target, the update

$$\lambda_j^{(t+1)} = \lambda_j^{(t)} \cdot \frac{\text{tr}(S_\lambda^+ S_j) - \text{tr}(V S_j)}{\hat{\beta}^T S_j \hat{\beta}}$$

is **monotone non-increasing** in $-V_r$ under mild conditions, has no line search, and converges geometrically near a minimum. The implementation lives at `src/fitting/solver.rs` (`run_optimization_fellner_schall`) and applies three safeguards:

Safeguard	Constant	Value	Role
Relative floor on the numerator $\text{tr}(S_\lambda^+ S_j) - \text{tr}(V S_j)$	<code>FS_NUMERATOR_FLOOR_REL</code>	10^{-12}	Prevents tiny / occasionally negative numerators from collapsing the update.
Floor on the denominator $\hat{\beta}^T S_j \hat{\beta}$	<code>FS_DENOMINATOR_FLOOR</code>	10^{-12}	Avoids division by zero when $\hat{\beta}$ lies in the null space of S_j .
Per-iteration clamp on $\log \lambda_j$	<code>LOG_LAMBDA_CLAMP</code>	± 30	Keeps λ in $[e^{-30}, e^{30}]$, matching the REML L-BFGS clamp.

Loop control: at most `FS_MAX_ITERS` = 50 iterations; converged when

$$\max_j |\log \lambda_j^{(t+1)} - \log \lambda_j^{(t)}| < \text{FS_TOL} = 10^{-4}.$$

The tolerance is set at 10^{-4} (rather than the looser 10^{-3}) because the F-S update is first-order convergent: stopping at 10^{-3} leaves λ still moving by $\approx 0.1\%$ per step, which is measurable on flat LAML landscapes. The extra iterations cost little since each F-S step is a single PWLS solve. Source: `src/fitting/solver.rs` (`FS_TOL`, `FS_MAX_ITERS`).

1.8.7.1 When to pick which criterion

- **GCV**: classical choice, computationally cheap; tends to undersmooth at moderate n and is more prone to multiple local minima on the GCV surface (Reiss & Ogden 2009). The GCV optimizer always runs L-BFGS from the warm-started λ and does not apply an early-exit threshold — a previous skip triggered when the warm-start GCV score was below 10^{-6} could freeze λ for parameters with small residual sums of squares (e.g. a log-scale σ whose working RSS is tiny), preventing those parameters from adapting their smoothness across outer cycles.
- **REML** (default): preferred for stability; the Laplace-approximate marginal likelihood is asymptotically equivalent to true REML for the working PWLS model, and tends to undersmooth less than GCV at moderate sample sizes.
- **FellnerSchall**: same target as REML but with a deterministic multiplicative update — no L-BFGS, no line search. Fast and well-behaved for well-conditioned problems; can stall if the numerator drifts to its floor.

1.8.8 8.4 Basin Probes (Collapse and Corner Guards)

For a **single** P-spline, the λ -objective ($-V_r$ or the GCV score) is unimodal in $\log \lambda$ with one interior optimum, but it flattens into a near-horizontal **shelf** at large λ where the smooth has been driven entirely onto its penalty null space ($\text{EDF} \rightarrow \text{null dimension}$; for an order-2 penalty after centering, a straight line). On the shelf the gradient ≈ 0 and, for Fellner-Schall, $\hat{\beta}^T S_j \hat{\beta} \rightarrow 0$ so the multiplicative ratio explodes upward — either optimizer can become stuck there.

For **multi-penalty** terms (the anisotropic tensor penalty $\lambda_1 S_1 \otimes I + \lambda_2 I \otimes S_2$) the surface is genuinely **multimodal**: spurious stationary points appear as *corners* — one margin's λ pinned at the clamp ceiling or the λ floor while the true optimum has it interior — and they come in shapes no cheap detector reliably catches (observed on real data: a corner with $\text{EDF} = 16.0$ scoring 5 LAML units worse than the interior optimum at $\text{EDF} = 20.6$, which matches `mgcv`'s selection). Gradient descent from any single seed can be captured by a basin boundary.

The selector in `src/fitting/scoring.rs::step` therefore probes alternative basins and keeps the λ with the best objective value (`lambda_cost`):

1. **Trigger.** Single-penalty terms probe when a smooth’s EDF has fallen to within `EDF_COLLAPSE_SLACK` = 0.5 of its null dimension, or when any λ sits at the log λ clamp bounds. Multi-penalty terms probe **unconditionally on every cycle** — their multimodality is intrinsic, and a “probe only when suspicious” gate was tried and lost real rescues (the early probes run against unconverged working weights; a skip-once-confirmed rule then never re-probes at the converged state where the rescue is decidable).
2. **Seeds.** (a) A low- λ restart seed $\exp(\log \lambda_{\text{cold}} - \text{RESTART_LOG_OFFSET})$ with offset 8, below both the interior optimum and the shelf; (b) a fresh cold start from the trace-ratio heuristic; (c) per-coordinate variants of the incumbent with each bound-pinned λ_j individually dropped to the restart level — the targeted escape for tensor corners; (d) for terms with ≤ 2 penalties, L-BFGS started from the best cell of a coarse 7^k grid of log λ offsets $\{-16, -12, \dots, +8\}$ around the cold start. The grid evaluation is derivative-free, so it cannot be captured by a basin boundary.
3. **Acceptance.** Every candidate (and the incumbent) is scored by the actual criterion value; the minimum wins. This makes the guard safe in both directions: a genuinely null-space-optimal fit (a strictly linear truth under an order-2 penalty) has the *better* marginal likelihood at the collapsed λ and is preserved; only a spuriously collapsed or corner-trapped fit — where another basin scores better — is repaired.

Source: `src/fitting/scoring.rs` (`step`), `src/fitting/solver.rs` (`restart_seed`, `lambda_cost`, `initial_log_lambda`, `RESTART_LOG_OFFSET`).

Determinism. One historical *trigger* for spurious single-smooth collapse is the nondeterministic reduction order of multi-threaded OpenBLAS. This repo pins BLAS to a single thread for its own cargo runs (`OPENBLAS_NUM_THREADS=1`, `OMP_NUM_THREADS=1` in `.cargo/config.toml`), making the dense linear algebra reproducible; under single-thread BLAS the recovery control cases collapse in 0/20 repeats (`tests/lambda_bistability.rs`). The basin probes above are retained as defense-in-depth for multi-threaded execution paths, and are the *primary* (not defensive) mechanism for multi-penalty terms.

1.9 9. Mean-Variance Relationships

A key feature distinguishing different distributions is how variance relates to the mean. This determines which distribution is appropriate for different data structures.

Distribution	Mean	Variance	Relationship
Gaussian	μ	σ^2	Constant (homoskedastic)
Poisson	μ	μ	Equidispersion: Var = Mean
Gamma	μ	$\mu^2 \sigma^2$	Proportional to μ^2 (constant CV)
Negative Binomial	μ	$\mu + \sigma \mu^2$	Overdispersion relative to Poisson
Binomial	$n\mu$	$n\mu(1 - \mu)$	Quadratic in mean, max at $\mu = 0.5$
Beta	μ	$\frac{\mu(1-\mu)}{1+\phi}$	Max at $\mu = 0.5$, decreases with ϕ
Student-t	μ	$\frac{\nu\sigma^2}{\nu-2}$	Heavier tails than Gaussian
Ordered Categorical	$\sum_r r \pi_r$	$\sum_r r^2 \pi_r - (\mathbb{E}Y)^2$	Discrete $\{1, \dots, R\}$; depends on all thresholds

1.9.1 Choosing a Distribution

Ordinal responses: - Fixed ordered categories $\{1, \dots, R\}$: Ordered Categorical (`ocat`)

Count data: - Equidispersed: Poisson - Overdispersed (variance $>$ mean): Negative Binomial - Known number of trials: Binomial

Continuous positive: - Constant coefficient of variation: Gamma - Heavy tails: Student-t with ν small - Heteroskedastic with mean: Gamma or lognormal (not implemented)

Proportions/rates in $(0, 1)$: - Beta (continuous) - Binomial (discrete counts / n trials)

Continuous unbounded: - Light tails, constant variance: Gaussian - Heavy tails: Student-t

1.9.2 Overdispersion

Overdispersion occurs when the observed variance exceeds what the distribution predicts from the mean alone.

For count data: - If $\text{Var}(Y) > \mu$: Use Negative Binomial instead of Poisson - The parameter σ in NB quantifies the overdispersion: as $\sigma \rightarrow 0$, NB $\rightarrow$ Poisson

For continuous data: - Gamma allows variance to scale with μ^2 - Student-t allows heavier tails (more extreme values) than Gaussian

1.10 10. Convergence and Initialization

1.10.1 Initialization Strategy

The RS algorithm requires starting values for all parameters. Default initialization:

1. μ :

- Gaussian: $\bar{y}$ (sample mean)
- Student-t: $\text{median}(y)$ — a robust location seed. The sample mean is pulled by the heavy tails, biasing the first robustifying weights $w = (\nu + 1)/(\nu + z^2)$.
- Poisson/Gamma/NB: $\bar{y}$ (then apply log link)
- Binomial: $\sum_i y_i / \sum_i n_i$ — pooled across observations so per-row trial counts do not bias the seed — clamped to $(0.1, 0.9)$
- Beta: $\bar{y}$ (sample mean, clamped to $(0.1, 0.9)$)

2. σ :

- Gaussian: s_y (sample std dev)
- Student-t: $1.4826 \cdot \text{MAD}(y)$ — the MAD-to- σ consistency factor for a normal core. A raw sample SD overestimates the scale under heavy tails. Floored at 10^{-4} (falling back to 1.0).
- Gamma: $\hat{\sigma}_0 = s_y / \bar{y}$ (sample CV), clamped to $[0.05, 10.0]$. σ parameterizes the coefficient of variation, not the raw SD; a raw-SD start makes REML over-penalize the σ smooth on the first RS iteration and warm-start into a full-collapse trap.
- NB: the method-of-moments overdispersion estimate $(s_y^2 - \bar{y})/\bar{y}^2$, clamped to $[0.1, 10.0]$ (falling back to 0.5 when undefined, e.g. $n = 1$). σ is the overdispersion coefficient of $\text{Var} = \mu + \sigma\mu^2$, so a raw-SD start is on the wrong scale entirely — often 10–30 $\times$ too large for count data.
- Beta: 1.0

3. ν (Student-t): 5.0 — a fixed moderate seed, deliberately **not** a sample-kurtosis estimate. For a regression model the *marginal* kurtosis of y reflects the spread of the mean structure rather than the noise tails, so inverting $\kappa = 6/(\nu - 4)$ biases ν ; in the multi-smooth weighted case this biased seed tipped the optimizer into a degenerate over-smoothed basin. 5 sits well clear of the $\nu > 2$ finite-variance boundary. Source: `StudentT::initial_value` in `src/distributions/student_t.rs`.

4. ϕ (Beta): 1.0

5. **Smoothing parameters** λ : initialized from the trace-ratio cold-start heuristic

$$\log \lambda_j^{(0)} = \log \left(\frac{\text{tr}(X^T X)}{\text{tr}(S_j)} \right)$$

per penalty (unweighted, using the assembled X for the current distribution parameter). The all-ones start $\lambda = 1$ can leave $X^T W X + S_\lambda$ near-singular for high-cardinality bases (e.g. $k = 20$) or prior-weighted models where $X^T W X$ is scaled by large prior weights. Source: `src/fitting/mod.rs`, `src/fitting/solver.rs` (`initial_log_lambda`).

1.10.2 Convergence Criterion

The algorithm declares convergence when every distribution parameter θ_k independently satisfies the relative criterion:

$$\frac{\|\beta_k^{(t+1)} - \beta_k^{(t)}\|_\infty}{\max(\|\beta_k^{(t+1)}\|_\infty, 1)} < \epsilon,$$

where k indexes parameters (μ , σ , etc.) and $\epsilon = 10^{-3}$ by default. Each parameter is checked against its **own** coefficient scale rather than a global scale pooled across all parameters. The $\max(\cdot, 1)$ floor makes the test behave like an absolute threshold when all coefficients are $O(1)$ (typical for normalized data).

Using a shared global scale — dividing the maximum change across all parameters by the maximum $|\beta|$ across all parameters — was the previous behaviour. It caused the loop to declare convergence prematurely when one parameter (e.g. μ) had large coefficients and dominated the denominator while another (e.g. log-scale σ) was still drifting in relative terms.

Maximum iterations: 200 (default)

1.10.3 Convergence Issues

Non-convergence can occur due to: 1. **Poor initialization:** Try better starting values, especially for ν in Student-t 2. **Model misspecification:** Wrong distribution family for the data 3. **Multicollinearity:** Highly correlated predictors (check condition number of design matrix) 4. **Insufficient smoothing:** Extremely small λ can cause instability 5. **Numerical issues:** Extreme data values causing overflow/underflow

Diagnostics: - Monitor deviance: should decrease monotonically - Check for NaN/Inf in coefficients or fitted values - Inspect condition number of $(X^T W X + \lambda S)$

1.11 11. Model Selection and Diagnostics

1.11.1 11.1 Information Criteria

Akaike Information Criterion (AIC):

$$\text{AIC} = -2\ell(\hat{\theta}) + 2k$$

where $\ell(\hat{\theta})$ is the maximized log-likelihood and k is the number of parameters.

For GAMLSS with smoothing:

$$k = \sum_{j=1}^P \text{EDF}_j$$

where P is the number of distribution parameters and EDF_j accounts for the effective degrees of freedom of smooth terms.

Bayesian Information Criterion (BIC):

$$\text{BIC} = -2\ell(\hat{\theta}) + k \log(n)$$

BIC penalizes complexity more heavily than AIC for $n > 7$.

1.11.2 11.2 Deviance

The **deviance** measures lack of fit:

$$D = 2[\ell_{\text{saturated}} - \ell_{\text{fitted}}]$$

where $\ell_{\text{saturated}}$ is the log-likelihood of a saturated model (perfect fit).

For nested models, the deviance difference follows approximately χ^2 distribution:

$$D_1 - D_2 \sim \chi^2_{\text{EDF}_2 - \text{EDF}_1}$$

1.11.3 11.3 Residuals

Quantile (randomized quantile) residuals (Dunn & Smyth 1996). For a **continuous** response:

$$r_i = \Phi^{-1}(F(y_i | \hat{\theta}_i))$$

where F is the fitted CDF and Φ^{-1} is the inverse standard normal CDF. If the model is correct, $r_i \sim N(0, 1)$ by the probability integral transform (§1.10).

For a **discrete** response, F jumps, so $F(Y)$ is not uniform; the *randomized* PIT spreads each atom across its jump interval. With $v_i \sim \text{Uniform}(0, 1)$:

$$a_i = F(y_i - 1 | \hat{\theta}_i), \quad b_i = F(y_i | \hat{\theta}_i), \quad u_i = a_i + v_i(b_i - a_i), \quad r_i = \Phi^{-1}(u_i).$$

The fitted CDF F , quantile Q , and the `is_discrete` flag that selects the branch are provided by the distributional trio of §1.10; the right-continuity identity $F(k) - F(k - 1) = f(k)$ is what makes u_i uniform on the jump interval.

Deviance residuals:

$$r_i^{(D)} = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}$$

where d_i is the contribution of observation i to the total deviance.

Pearson residuals:

$$r_i^{(P)} = \frac{y_i - \hat{\mu}_i}{\sqrt{\widehat{\text{Var}}(Y_i)}}$$

1.11.4 11.4 Worm Plots

For each parameter θ_k , plot quantile residuals against fitted quantiles: - X-axis: Normal quantiles $\Phi^{-1}((i - 0.5)/n)$ - Y-axis: Sorted quantile residuals

If the model is correct, points should lie on a horizontal line at zero. Systematic deviations indicate: - U-shape: Underdispersion - Inverse U-shape: Overdispersion - S-shape: Skewness issues - Linear trend: Location shift

1.11.5 11.5 Q-Q Plots

Plot theoretical quantiles against sample quantiles of residuals. Should be approximately linear if residuals are normal.

1.11.6 11.6 Goodness of Link Test

To test if link g is appropriate, fit augmented model:

$$g(\theta) = X\beta + \gamma[g(\theta)]^2$$

Test $H_0 : \gamma = 0$ using likelihood ratio test.

1.12 12. Code Map

A reverse index from mathematical concept to the canonical implementation, for contributors jumping between formula and source.

Concept (this doc)	Source location
Backfitting outer loop with per-parameter convergence (§4.1)	src/fitting/mod.rs (fit_gamlss)
P-IRLS inner step (§4.2)	src/fitting/scoring.rs (step)
MIN_WEIGHT, MAX_STEP (§4.4)	src/fitting/scoring.rs,32
Prior / observation weights (§4.5)	src/fitting/scoring.rs (step)
MAX_ETA, MIN_ETA, MIN_POSITIVE (§4.4)	src/distributions/links.rs
IdentityLink, LogLink, LogitLink (§7)	src/distributions/links.rs
Gaussian derivatives (§1.1)	src/distributions/gaussian.rs
Student-t derivatives (§1.2)	src/distributions/student_t.rs
Poisson derivatives (§1.3)	src/distributions/poisson.rs
Gamma derivatives (§1.4)	src/distributions/gamma.rs
Negative Binomial derivatives (§1.5)	src/distributions/negative_binomial.rs
Beta derivatives (§1.6)	src/distributions/beta.rs
Binomial derivatives (§1.7)	src/distributions/binomial.rs
Ordered categorical derivatives (§1.8)	src/distributions/ocat.rs
Digamma / trigamma batched (§2)	src/math.rs
Distribution trait (§1)	src/distributions/mod.rs
CDF / PDF / quantile / is_discrete trait methods (§1.10)	src/distributions/mod.rs + per-family files
discrete_quantile bracket+bisection (§1.10)	src/distributions/mod.rs (discrete_quantile)
B-spline basis (de Boor) & uniform knots (§5.0)	src/splines/pspline.rs (create_basis_matrix_with_range, evaluate_basis_functions_into, select_knots)
Sum-to-zero Householder Z (§5.1)	src/splines/reparam.rs (sum_to_zero_basis)
Difference penalty matrix (§5)	src/splines/ (create_penalty_matrix)
Tensor product assembly (§5)	src/fitting/assembler.rs
Random effect assembly (§5)	src/fitting/assembler.rs
Block-sparse penalty (§5.1)	src/fitting/assembler.rs, src/types/newtypes.rs
PWLS / Cholesky solve (§8.1)	src/fitting/solver.rs (fit_pwls, fit_pwls_with_grad_info)
Robust log-det fallback (§8.1, §8.2)	src/linalg.rs (log_det_robust)
Initial- λ trace-ratio heuristic (§10)	src/fitting/mod.rs, src/fitting/solver.rs (initial_log_lambda)
EDF (§6)	src/fitting/solver.rs (fit_pwls_with_grad_info)
GCV cost and gradient (§8)	src/fitting/solver.rs (GamlssCost)

Concept (this doc)	Source location
GCV gradient finite-difference test	src/fitting/solver.rs (gcv_gradient_matches_finite_diff)
GCV L-BFGS driver (§8)	src/fitting/solver.rs (run_optimization)
REML / LAML cost and gradient (§8.2)	src/fitting/solver.rs (RemlCost)
REML gradient finite-difference test	src/fitting/solver.rs (reml_gradient_matches_finite_diff)
Penalty grouped pseudo-determinant / pseudo-inverse (§8.2)	src/fitting/solver.rs (penalty_eigen, penalty_nonzero_block_range)
Fellner–Schall iteration (§8.3)	src/fitting/solver.rs (run_optimization_fellner_schall)
SmoothingCriterion enum (§8.2)	src/fitting/mod.rs
Defaults (max_iterations=200, tolerance=1e-3)	src/fitting/mod.rs
Diagnostics (AIC/BIC/EDF/residuals, §11)	src/fitting/diagnostics.rs

1.13 13. Notation Glossary

Symbols used throughout this document.

Symbol	Meaning
y, Y	Response value / random variable
n	Number of observations
p	Total number of coefficients across all terms; also (in §5.0) B-spline degree
P	Number of distribution parameters (e.g. $P = 2$ for Gaussian)
θ_k	k -th distribution parameter (μ, σ, ν, ϕ)
μ, σ, ν, ϕ	Location, scale, degrees of freedom (Student-t), precision (Beta)
R	Number of ordered categories in <code>0cat</code> ($R \in \{2, 3, 4, 5\}$)
θ_k	k -th threshold in <code>0cat</code> ($k = 1, \dots, R - 1$); also used generically for the k -th distribution parameter — context distinguishes
δ_k	k -th threshold increment parameter fed to the RS loop: $\theta_1 = \delta_1$ (identity link), $\theta_k = \theta_{k-1} + e^{\delta_k}$ for $k \geq 2$ (log link)
F_k, f_k	Cumulative-logit CDF and logistic density at threshold θ_k : $F_k = \text{logistic}(\theta_k - \mu)$, $f_k = F_k(1 - F_k)$
π_r	Category probability in <code>0cat</code> : $\pi_r = F_r - F_{r-1}$, $F_0 = 0$, $F_R = 1$
$F(y \mid \theta)$	Cumulative distribution $P(Y \leq y \mid \theta)$; right-continuous step at $\lfloor y \rfloor$ for discrete families (§1.10)
$f(y \mid \theta)$	Density (continuous) or mass (discrete) $= \exp(\ell_{\text{pointwise}})$ (§1.10)
$Q(p \mid \theta)$	Quantile / inverse CDF: $\inf\{y : F(y \mid \theta) \geq p\}$ (§1.10)
$P(a, x)$, $Q(a, x)$	Regularized lower / upper incomplete gamma $\gamma(a, x)/\Gamma(a)$, $1 - P(a, x)$
$I_x(a, b)$	Regularized incomplete beta $B(x; a, b)/B(a, b)$

Symbol	Meaning
Φ, Φ^{-1}	Standard normal CDF and its inverse (quantile)
prior_i	Per-observation prior / observation weight (§4.5); defaults to 1
safe_w_i	Floored Fisher weight: $\max(w_i, \text{MIN_WEIGHT})$
g, g^{-1}	Link function and its inverse
$\eta = g(\theta)$	Linear predictor for a distribution parameter
X	Design matrix ($n \times p$); collects intercept, linear, and smooth basis columns
β	Coefficient vector ($p \times 1$)
W	Diagonal weight matrix; $W_{ii} = w_i = \text{Fisher info for observation } i$
$u_i = \partial \ell / \partial \eta_i$	Score (first derivative of log-likelihood on η -scale)
$w_i = -\mathbb{E}[\partial^2 \ell / \partial \eta_i^2]$	Fisher information on η -scale
$z = \eta + u/w$	Working response
S_j	j -th penalty matrix (one per smooth term direction)
λ_j	Smoothing parameter for S_j
$S_\lambda = \sum_j \lambda_j S_j$	Aggregate penalty matrix
$V = (X^T W X + S_\lambda)^{-1}$	PWLS coefficient covariance
$H = X V X^T W$	Hat matrix (projection onto fitted values)
$\text{EDF} = \text{tr}(H) = \text{tr}(V X^T W X)$	Effective degrees of freedom
$\text{RSS} = (z - X \hat{\beta})^T W (z - X \hat{\beta})$	Weighted residual sum of squares
$\psi(x), \psi'(x)$	Digamma, trigamma
$ A _+$	Pseudo-determinant of PSD matrix A (product of strictly positive eigenvalues)
A^+	Moore–Penrose pseudo-inverse of A
$\otimes$	Kronecker product
$\odot$ (in §5)	Row-wise Kronecker product (one row of $B_1 \otimes B_2$ per row index)
$\mathbf{1}_k$	Column vector of k ones
I_k	$k \times k$ identity matrix

1.14 References

1. Rigby, R. A., & Stasinopoulos, D. M. (2005). Generalized additive models for location, scale and shape. *Journal of the Royal Statistical Society: Series C*, 54(3), 507-554. — Core RS backfitting algorithm and observed-info choice in NB.
2. Eilers, P. H., & Marx, B. D. (1996). Flexible smoothing with B-splines and penalties. *Statistical Science*, 11(2), 89-121. — P-spline difference penalty.
3. de Boor, C. (1978). *A Practical Guide to Splines*. Springer. — Cox–de Boor B-spline recursion.
4. Wood, S. N. (2017). *Generalized Additive Models: An Introduction with R* (2nd ed.). Chapman and Hall/CRC. — Tensor-product smooths, identifiability constraints, GAM reference.
5. Wood, S. N. (2011). Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models. *Journal of the Royal Statistical Society: Series B*, 73(1), 3-36. — Laplace-approximate REML for smoothing selection.
6. Wood, S. N., & Fasiolo, M. (2017). A generalized Fellner–Schall method for smoothing parameter optimization with application to Tweedie location, scale and shape models. *Biometrics*, 73(4), 1071-1081. — Multiplicative fixed-point update.

7. Reiss, P. T., & Ogden, R. T. (2009). Smoothing parameter selection for a class of semiparametric linear models. *JRSS B*, 71(2), 505-523. — Comparison of GCV vs REML behavior.
8. Craven, P., & Wahba, G. (1979). Smoothing noisy data with spline functions. *Numerische Mathematik*, 31, 377-403. — Original GCV.
9. McCullagh, P., & Nelder, J. A. (1989). *Generalized Linear Models* (2nd ed.). Chapman and Hall. — IRLS / Fisher scoring derivation underlying §4.3.
10. Abramowitz, M., & Stegun, I. A. (1972). *Handbook of Mathematical Functions*. Dover Publications. — Digamma / trigamma asymptotic expansions (6.3.18, 6.4.11).
11. Piegl, L., & Tiller, W. (1997). *The NURBS Book* (2nd ed.). Springer. — Algorithm A2.2: de Boor–Cox recurrence for stable B-spline evaluation; the canonical reference for the `left/right` array structure used in `evaluate_basis_functions_into`.
12. McCullagh, P. (1980). Regression models for ordinal data. *Journal of the Royal Statistical Society: Series B*, 42(2), 109-142. — Proportional-odds / cumulative-logit model; foundational reference for the `Ocat` distribution (§1.8).
13. Dunn, P. K., & Smyth, G. K. (1996). Randomized quantile residuals. *Journal of Computational and Graphical Statistics*, 5(3), 236-244. — Randomized probability integral transform for discrete responses (§1.10, §11.3).